A Probabilistic Analysis of the Conditioning of Algebraic Rootfinding Methods

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Abstract

To be written.

1 Introduction

It was shown in [9, 7] that all known algebraic methods for numerically finding a solution to a system of polynomial equations $p_1(x) = \cdots = p_d(x) = 0$, $x \in \mathbb{C}^d$, are exponentially unstable: the condition number of the subproblems involved — specifically, eigenvalue or univariate polynomial rootfinding problems — can have exponentially (in d) worse conditioning than the original problem. These results, however, rest on carefully designed perfidious examples, akin to the Wilkinson polynomial in the univariate case [11]. In this article, we investigate whether this deterioration is a typical phenomenon or a pathology of the examples, by subjecting the methods to a probabilistic analysis. The probabilistic analysis of conditioning has been central to another popular class of rootfinding methods, homotopy continuation methods [10, 4, 6], where it plays a key role in the solution of Smale's 17th problem [2, 5, 8]. The present paper brings this average-case viewpoint to the algebraic rootfinders.

The methods considered in this paper can be grouped into three categories: hidden-variable resultant methods — Cayley and Sylvester resultants, multiparameter eigenvalue problems (MEP) —, scalar elimination methods — Gröbner elimination, rational univariate represantation (RUP) —, and normal form methods.

Let $P = (p_1, \dots, p_d) \colon \mathbb{C}^d \to \mathbb{C}^d$ be a polynomial map with N simple roots $z_1, \dots, z_N \in \mathbb{C}^d$ (that is, roots for which the Jacobian $J(z_i) := \mathrm{d}P(z_i)$ has rank d). The *intrinsic condition number* of a root z is defined as

$$\kappa_{\mathrm{root}}(z) = \|J(z)^{-1}\| = \sigma_{\min}(J(z))^{-1},$$

where $\|\cdot\|$ is the Euclidean norm.

1.1 Notation

For a multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$, we write $x^\alpha := x_1^{\alpha_1} \cdots x_d^{\alpha_d}$. We denote by $\mathbb{C}\mathcal{N}(0,1)$ the complex normal distribution, where real and imaginary parts are independent and have variance $\frac{1}{2}$.

2 Probabilistic background

Let d and n be positive integers. For $i \in \{1, \dots, d\}$, we consider random polynomials of the form

$$p_i(x) = \sum_{\alpha \in \{0,\dots,n\}} c_{i,\alpha} x^\alpha, \quad c_{i,\alpha} \sim \mathbb{C}\mathcal{N}(0,1). \tag{1}$$

The space of coefficients $c_{i,\alpha}$ is naturally identified with $\mathcal{V} := \mathbb{C}^d \times (\mathbb{C}^{n+1})^{\otimes d}$ and the random coefficients define a $\mathcal{V}$-valued random variable P. There are practical and conceptual reason for using the maximal-degree model of [9], rather than taking n to be the total degree. On the one hand, it is motivated by its link to tensor-product constructions [9, Sec. 2.1]. A further reason is that fixing the maximal degree in each variable keeps the Newton polytopes - the convex hull of the exponent vectors - equal to boxes, for which the resultant constructions of Section **??** are generically nonsingular and there are generically no roots at infinity. Each p_i is a Gaussian random function with covariance kernel

$$K(x,y) := \mathbb{E}\big[p_i(x)\overline{p_i(y)}\big] = \prod_{\ell=1}^{d} k_n(x_\ell \bar{y}_\ell), \qquad k_n(t) := \sum_{j=0}^{n} t^j, \tag{2}$$

and the d functions are independent. Write $S(X) := K(X,X) \geq 1$ and $r_\ell = |X_\ell|^2$.

By the Bernstein–Kushnirenko–Khovanskii (BKK) theorem [3], using the normalization $\mathrm{MV}(Q, \dots, Q) = d!\,\mathrm{Vol}(Q)$ for the mixed volumes (Q a polytope), a generic system with exponent vectors in the box $[0,n]^d$ has

$$N = d!\ \mathrm{Vol}([0,n]^d) = d!\, n^d \tag{3}$$

simple roots in the torus $(\mathbb{C}^*)^d$. Moreover, the set of coefficients for which equality is not attained is a proper algebraic subset of $\mathcal{V}$, so almost surely all N roots are simple. For such a generic system, for which all N roots are simple, the ideal $I = \langle p_1, \dots, p_d \rangle$ is radical and zero-dimensional, and evaluation at the roots identifies $\mathbb{C}[x]/I$ with $\mathbb{C}^N$.

2.1 The Kac-Rice formula

The main probabilistic tool in our analysis is a corollary of the Kac-Rice formula. Let $\{X(t) \colon t \in U\}$ be a random field indexed by an open set $U \subseteq \mathbb{R}^d$. For $u \in \mathbb{R}^d$ and $S \subseteq U$, define the counting function

$$N_u(X; S) = \#\{t \in S : X(t) = u\} \in \mathbb{N} \cup \{\infty\}.$$

If X is C^1 a.s. and u is a regular value a.s., then $N_u(X;S)$ is locally finite a.s., and finite if S is compact.

Theorem 2.1 (Kac-Rice formula [1]) *Let $U \subseteq \mathbb{R}^d$ be an open set, and let $X\colon U \to \mathbb{R}^d$ be a non-degenerate Gaussian random field such that almost surely the trajectories $t \mapsto X(t)$ are C^1. Let $u \in \mathbb{R}^d$ be a point such that almost surely, u is a regular value of X, that is,*

$$\mathbb{P}\{\exists t \in U\colon X(t) = u \text{ and } \det(dX(t)) = 0\} = 0.$$

Then

A random polynomial system P can be seen as a random field.

Lemma 2.1 *Let $\varphi\colon \mathbb{C}^d \times \mathcal{V} \to \mathbb{R}$ be measurable. If*

$$\int_{\mathbb{C}^d} \mathbb{E}\big[\,|\varphi(x;P)|\,|\det J(x)|^2 \;\big|\; P(x) = 0\,\big]\, S(x)^{-d}\, dx < \infty, \tag{4}$$

then

$$\mathbb{E}\Big[\sum_{z\colon P(z)=0} \varphi(z;P)\Big] = \int_{\mathbb{C}^d} \mathbb{E}\big[\varphi(X;P)\,|\det J(X)|^2 \;\big|\; P(X) = 0\big]\, p_{P(X)}(0)\, dX, \tag{5}$$

holds with both sides finite. For $\varphi \geq 0$, (5) holds unconditionally, with both sides in $[0,\infty]$.

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